Optical fibers

An optical fibre is a thin stand of glass or plastic that serves as the transmission medium over which the information passes through

Advantages of optical fibre cables over coppa cables.

	parameter	fibre optics	coppor
1)	Bandwidth	60 Tbps & beyond	10Gbps
2)	Distance	12 miles + @ 10,000 Mbps	300 Ft. @ 1000 Mbps
3)	noise	immune	Susaptible to interference, cross talk & voltage surgy
4)	Security	Nearly impossible to tap	susaptible to tapping
5)	Handling	light weight thin diameter strong pulling strength	Heavy thicker diameter strict pulling specifications.
6)	Life cycle	30-50 years	5 years
7)	wt / 1000 ft.	4 Lbs	39 Lbs.
8)	Energy consumed	2W per user	>10W per user.

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For communication systems three parts are important

(1) Transmitter

(2) Transmission channel

(3) Receiver.

Transmission channel must transmits the signals without loss of energy (attenuation & distortion)

optical fiber

It is a hair thin cylindrical fiber of glass or any transparent dielectric medium.

In practical applications, they consists of many thousands of very long fine quality glass / quartz fibers.

Structure of optical fibers

An optical fiber consists three parts.

(1) The core

(2) The cladding

(3) The outer jacket.

(1) **The core**: Inner cylindrical part made of glass

Its diameter can vary from 5 μm to 100 μm.

It carries light.

(2) **The cladding**:

The core is sorrounded by another cylindrical shell of lower refractive index than core.

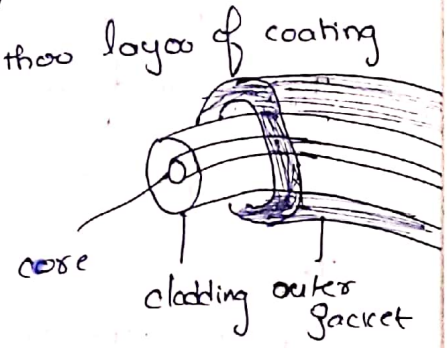
is called cladding.

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- It helps to keep the light within the core through the phenomenon of total internal reflection.
- Its diameter is 125 μm.

(3) outer jacket :

For better strength and protection of the fiber a soft plastic coating (primary coating) is done whose diameter is about 250 μm. This is often followed by another layer of coating known as secondary coating.



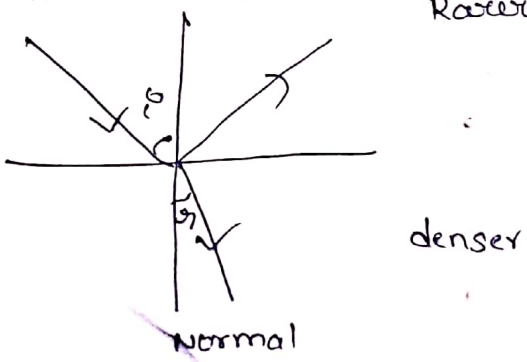
principle of optical fibers

optical fibers mainly works on the basis of total internal reflection.

Rarer medium & Denser medium

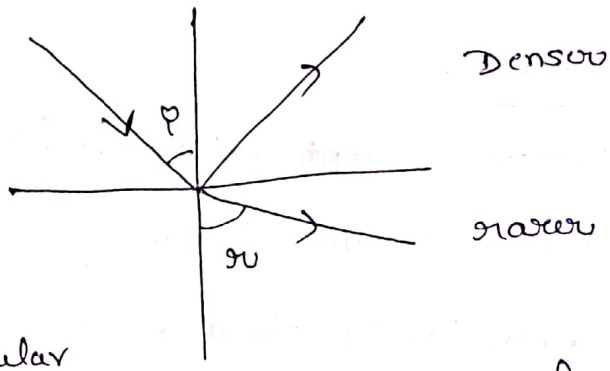
If we consider refractive indices of two materials. whose refractive index is low, that medium is called rarer medium & whose refractive index is high that medium is known as denser medium.

If light propagates from rarer to denser it goes towards the ~~from~~ normal



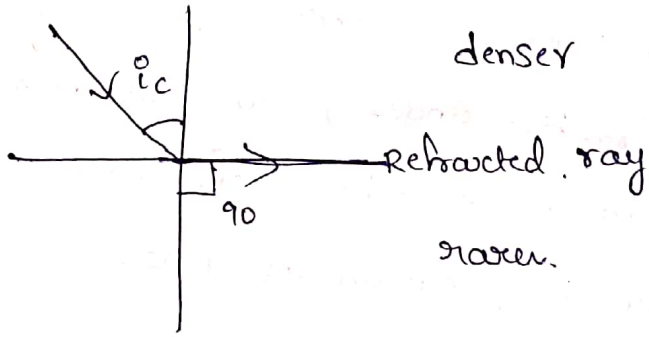
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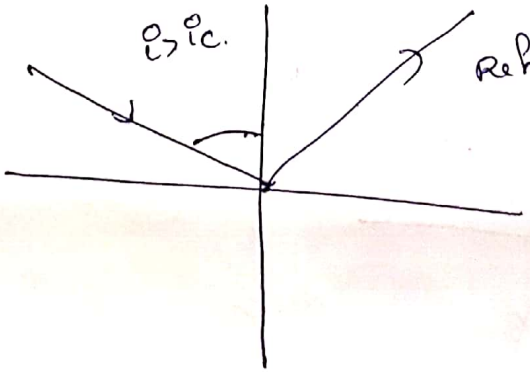
when the light propagates from denser to rarer the refracted light goes away from normal



At one particular angle of incidence i, the angle of refraction becomes 90°, that angle of incidence is called critical angle.

If incident ray makes an angle more than critical angle then, refracted ray comes into denser medium then, complete reflection takes place. This is called total internal reflection





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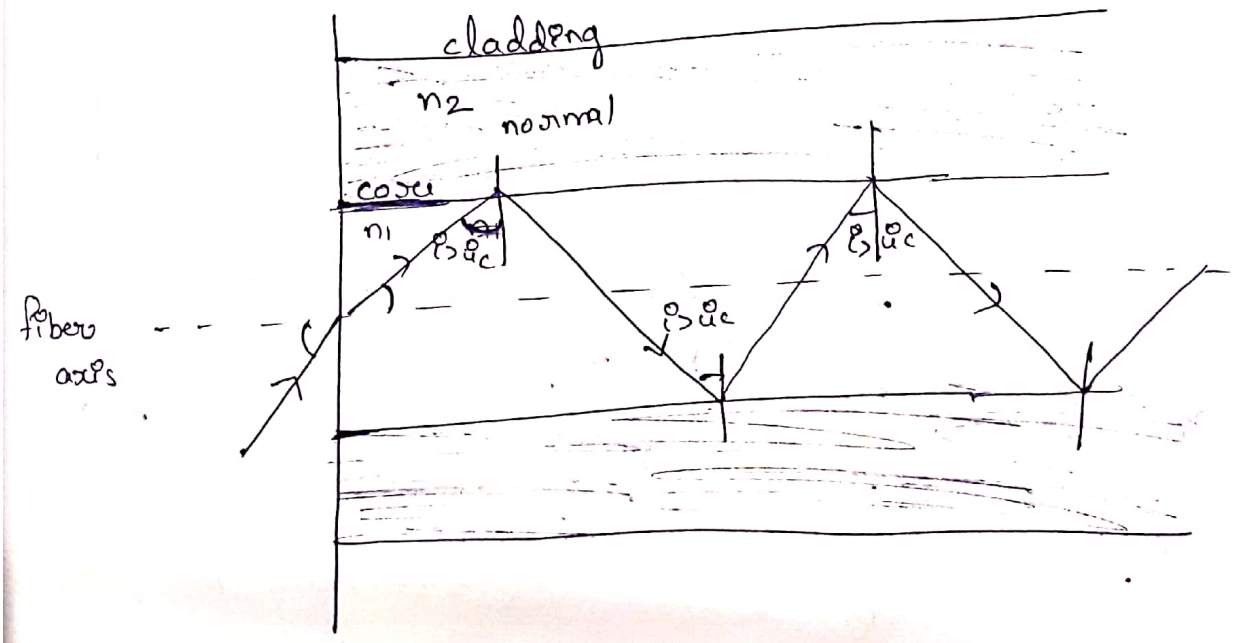
Total internal reflection at the fiber can occur only if it satisfies the following two conditions.

(1) The refractive index of core (n_1) must be slightly highest than the cladding. (n_2)

(2) At the core-cladding interface the angle of incidence θ (b/w the ray & the normal to the ~~surface~~ interface) must be greater than critical angle.

- when light travels from core-cladding (denser-rarer) the angle of refraction (r) is greater than the angle of incidence. (i)
- with the increase in 'i', r also increases
- If $i > i_c$, r becomes more & refracted ray, reflected back into the core at the interface obeying laws of reflection. This phenomenone is known as Total internal reflection.

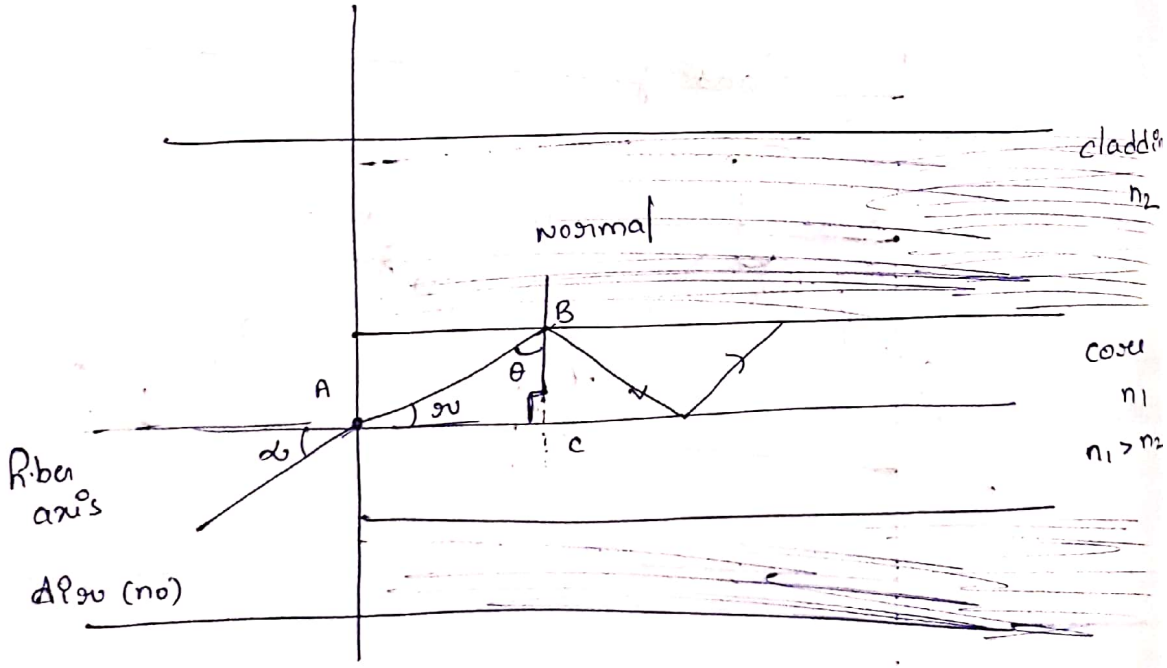
$n_1 > n_2$



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Acceptance angle & Acceptance cone

- when we launch the light beam into a fibre at its one end using focussing lens, the entire light may not pass through the core and propagate
- only the rays which make the angle of incidence greater than critical angle at the core-cladding interface can undergo total internal reflection and propagate through the core.
- The other rays are refracted to the cladding and are lost.
- Hence it is very essential to know ~~at~~ upto what angle one have to launch the beam at its end to enable the entire light to pass through the core. This maximum angle of launch is called acceptance angle.



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The light enters into the fiber with an angle 'α'.

r → refracted ray enters into fiber with an angle.

θ → angle of incidence at core-cladding interface.

From ΔABC

$$r + \theta + 90 = 180$$

$$r + \theta = 90$$

$$r = 90 - \theta \longrightarrow (1)$$

From snell's law. $\frac{\sin i}{\sin r} = \mu$ = refracting index. (Rarer → denser μ: Md/ger; denser-Rarer $\mu = \mu_r/\mu_d$)

$$\frac{\sin\alpha}{\sin r} = \frac{n_1}{n_0}$$ (air to core, $n_0 \rightarrow n_1$).

$$\sin\alpha = \frac{n_1}{n_0} \sin r$$

$$\sin\alpha = \frac{n_1}{n_0} \sin(90 - \theta) \quad (\text{from } (1))$$

$$\sin\alpha = \frac{n_1}{n_0} \cos\theta \longrightarrow (2)$$

If angle of incidence at core-cladding interface is more than critical angle then total internal reflection takes place otherwise not.

Hence at $\theta = \theta_c$ α becomes max angle that can gather max. light.

$$\sin(\alpha_{max}) = \frac{n_1}{n_0} \cos\theta_c \longrightarrow (3)$$

Snell's law at core-cladding interface. $(n_1 - n_2)$

$$\frac{\sin\theta_c}{\sin 90} = \frac{n_2}{n_1}$$ (at $i = \theta_c$, $r = 90°$)

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$$\sin\theta_c = \frac{n_2}{n_1} \rightarrow (4)$$

from (3)

$$\sin(\alpha_{max}) = \frac{n_1}{n_0}\cos\theta_c$$

$$\sin(\alpha_{max}) = \frac{n_1}{n_0}\sqrt{1-\sin^2\theta_c}$$

$$\sin[\alpha_{max}] = \frac{n_1}{n_0}\sqrt{1-\frac{n_2^2}{n_1^2}} \quad (\text{from (4)})$$

$$\sin(\alpha_{max}) = \frac{n_1}{n_0}\sqrt{\frac{n_1^2-n_2^2}{n_1^2}}$$

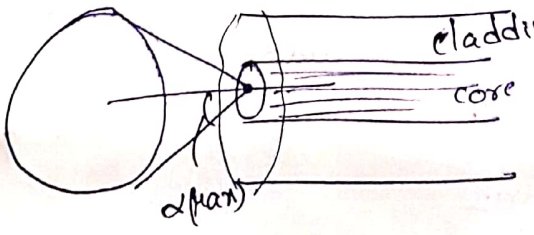
$$\sin(\alpha_{max}) = \frac{\cancel{n_1}}{n_0}\frac{\sqrt{n_1^2-n_2^2}}{\cancel{n_1}}$$

$$\sin(\alpha_{max}) = \frac{\sqrt{n_1^2-n_2^2}}{n_0}$$

$$\therefore \alpha_{max} = \sin^{-1}\left[\frac{\sqrt{n_1^2-n_2^2}}{n_0}\right]$$

This is the maximum acceptance angle.

Rotating the acceptance angle about the fiber axis is acceptance cone of the fiber.



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The light launched within this acceptance cone alone will be accepted and propagated to the other end of the fiber by total internal reflection.

Larger acceptance angles make launching easier.

Numerical Aperture (NA):

Light collecting capacity of the fiber is expressed interms of acceptance angle using the terminology 'Numerical aperture'.

Sine of the maximum acceptance angle is called the numerical aperture (NA) of the fiber.

$$NA = \sin\alpha(\text{Max}) = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

~~Let $\Delta = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$ for our convenience.~~

Let $\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$

for most of the fibers

$$n_1 \simeq n_2$$

$$\Delta = \frac{(n_1 - n_2)(n_1 + n_2)}{2n_1^2}$$

$$\Delta = \frac{(n_1 - n_2)(n_1 + n_1)}{2n_1^2} = \frac{(n_1 - n_2)(2n_1)}{2n_1^2}$$

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$$\Delta = \frac{n_1 - n_2}{n_1}$$

= Fractional difference b/w the refractive indices of core and cladding

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$$

$$\Delta (2n_1^2) = n_1^2 - n_2^2$$

$$\sqrt{2\Delta n_1^2} = \sqrt{n_1^2 - n_2^2}$$

$$n_1\sqrt{2\Delta} = \sqrt{n_1^2 - n_2^2}$$

$$NA = \sin(\alpha Max) = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

$$NA = \frac{n_1\sqrt{2\Delta}}{n_0}$$

If we launch the light into the fibre.

from air $n_0 = 1$ Then

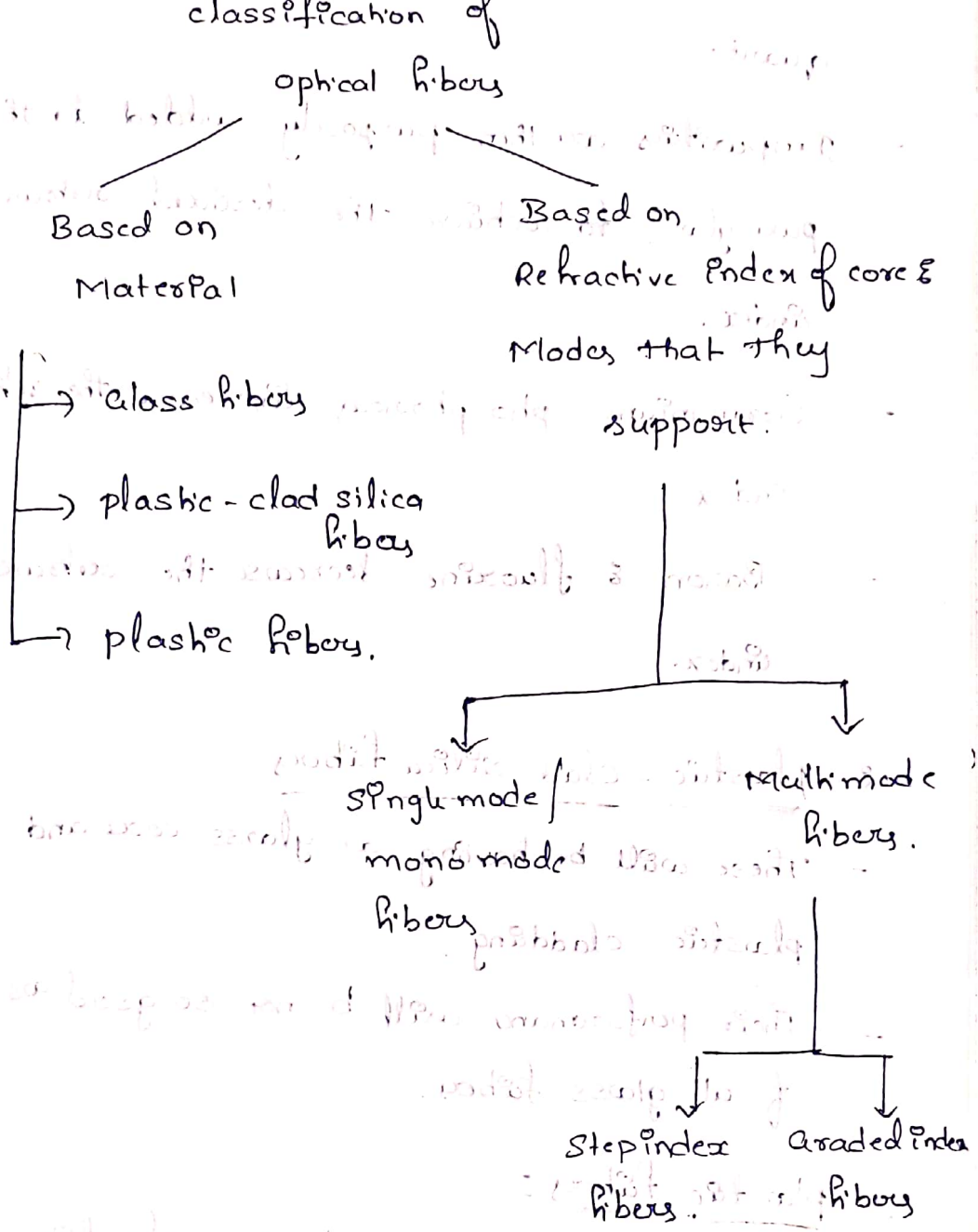
$$NA = n_1\sqrt{2\Delta}$$

NA mainly depends upon refractive indices of core and cladding of materials

It does not depends upon dimensions of optical fibre.

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Types of optical fibers



classification of fibers based on materials:

(1) Glass fibers

- These fibers will be having a glass core and also have glass cladding.
- These fibers are very widely used.

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- The glass used should be ultra pure and ultra transparent silicon dioxide or fused quartz.
- Impurities are then purposely added to this pure glass to achieve the desired refractive index.
- Germanium, phosphorous increase the refractive index
- Boron & fluorine decrease the refractive index.

2. plastic - clad silica fibers
- These will be having a glass core and a plastic cladding.
- Their performance will be not so good as that of all glass fibers.

3. plastic fibers:
- These optical fibers are having plastic core and plastic cladding.
- These are comparatively limited in losses and band width.
- Their very low cost and easy used, make them attractive in applications where large bandwidth

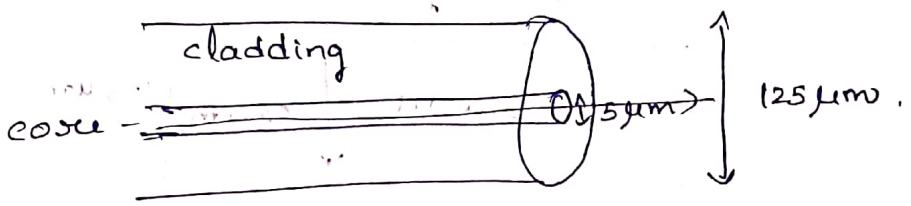
or loss losses are not required.

- They have got the advantages of electromagnetic immunity and security.

classification of fibers based on Refractive index & Modes of propagation

Single mode / Mono mode fibers:

- These will be having a very narrow core of diameter 5 μm or less.
- The size of the cladding will be very big. (125 μm)
- Because of very narrow core, only a single mode can propagate parallel to central axis through this type of fiber.



Multimode fibers:

- These fibers are having relatively large diameter of about 50μm.
- A large no. of modes (upto 100) can propagate through these fibers.
- Base upon refractive index of core these fibers are divided into → step index & two types → Graded index fiber

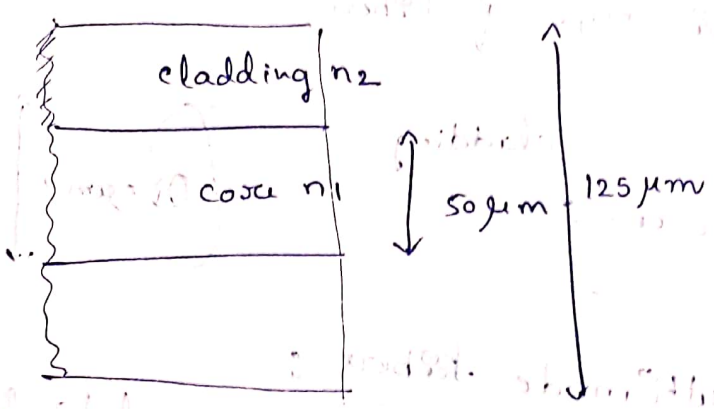
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Step Index fiber

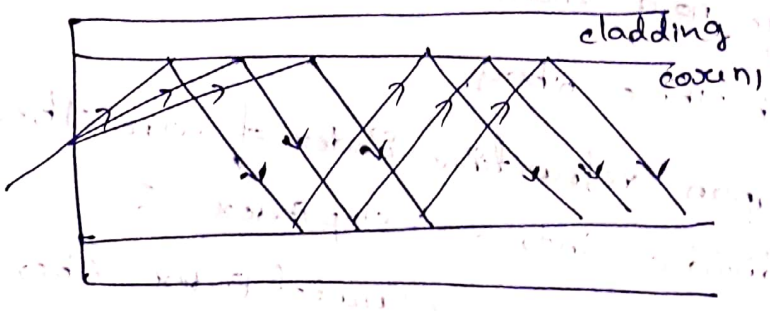
In this type the core will be relatively wide and having a constant refractive index (n_1) through out from its centre upto the boundary with the cladding.

Afterwards, the refractive changes to a constant lower value (n_2) inside the cladding.

This is called as step index multimode fiber because the refractive index (steps) down from n_1 to n_2 ($n_1 > n_2$) at the boundary with the cladding.



Transmission of signal in step index fiber



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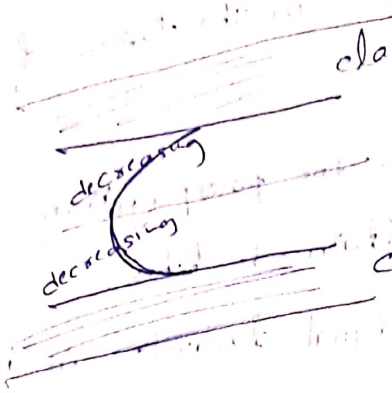
- Generally the signal is sent through the fiber in digital form i.e in the form of pulses such as 0s and 1s.
- let us now consider the propagation of one such pulse through multimode fiber.
- The same pulsed signal travels in different paths.
- Hence at the receiving end only the ray which travels along the fiber axis reaches first while other rays zig zag paths reach after some time delay.
- Hence the pulsed signal received at the other end broadened i.e called inter modal dispersion (pulse broadening)
- Due to this the transmission rate & capacity reduces.
- This can be overcome by graded index fiber.

Graded index fiber :-

In Graded index fiber, the refractive index of the core varies (decreases) smoothly & continuously in a nearly parabolic manner from the center of the core to the boundary of core-cladding.

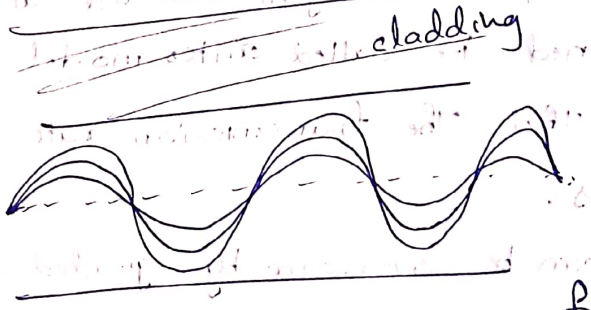
cladding's refractive index (n_2) is constant & is less than the core's refractive index (n_1)

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Transmission of signal in graded index fabor

In graded index fabor, the incident light consists of three diff. wavelengths.



- As the refractive index decreases from center of core to cladding, the ray entering the fibre is continuously bent towards the axis of fibre.
- This happens when the ray travels from core - cladding & cladding - core also.
- Though it travels longer distance, it travels along the path possessing relatively lower refractive index.

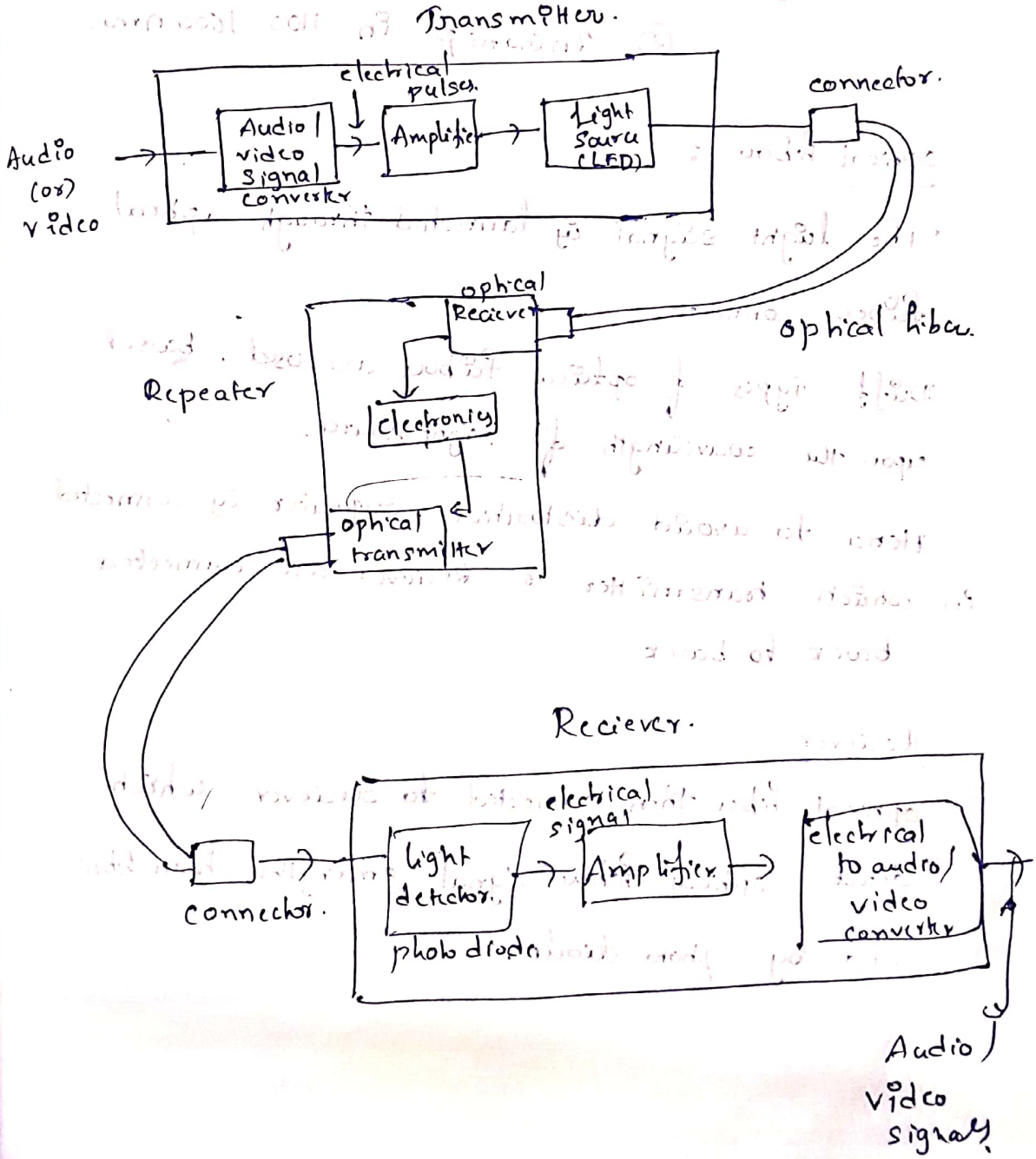
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- Hence all the rays comes out of fiber at the same time.
- Hence distortion of signal is reduced.

Optical fibre communication with Block Diagram

An optical fiber transmission link consists of mainly three parts.

(1) Transmitter section (2) optical fiber (3) Reciever.



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Transmitter:

- It consists of suitable electronic circuit that converts audio / video input to electric signals.
- The electrical signals are amplified & fed to light source i.e LED converts electrical signal to optical signal.
- Light sources are used based upon on light wavelength ① GaAlAs in 800-900nm
② InGaAsP in 1100-1600 nm.

Optical fibre:

The light signal is launched through optical fibre connectors.

Diff types of optical fibres are used based upon the wavelength of light source.

Hence to avoid distortion repeater is connected in which transmitter & Reciever are connected back to back

Reciever:

optical fibre then connected to reciever which detects optical ~~fibre~~ signal emerging from fibre end by photo diode.

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- photo diode converts the light signal to electrical signal.

Two Types of detectors (photodiodes) are used based upon received signal

① Semiconductor positive Intrinsic Negative (PIN) diode. for low power optical signal (800-900 nm)

② Avalanche photodiodes (APDs) (InGaAs alloy) diode (1100-1600 nm).

- The electrical signal is amplified again converted audio/video signal by converter.

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Optical fibers

1) Explain principle of optical fibers.
2) Discuss the construction of optical fiber. (by mentioning their dimensions of various parts)
3) Numerical aperture of an optical fiber.
4) Explain the process of transmission of signal through step index fiber, graded index fiber.
5) Explain fiber optic communication. (with block diagram)
6) Acceptance angle of an fiber.
7) Differentiate single and multimode fibers.
8) calculate the angle of acceptance of a fiber if RI of core & cladding are 1.563 & 1.498

~~Final~~ Model-2

1) Diff. b/w copper cables & optical fiber cables.
2) phenomenone of light denser to rarer
3) Diff. modes of propagation in an fiber (single & multimode)
4) NA problem
5) Detectors of fiber optical communication
6) few sources used as transmitters in fiber optic communication

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